

Diffused Photon Near-Infrared Tomography for Traumatic Brain Injury due to Concussions

System Architecture & Technical Design Specification

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Prepared by:

Ryan Vo

Thomas Molnar

Nathan Huynh

Abstract

The goal is to design a fNIR system that can detect a TBI. To accomplish this, the design has a transmitter shoot laser into the brain while having a receiver detect the photons and output bits into a microcontroller that can be wirelessly transmitted to a laptop where calculations and a 3D image of the brain can be generated. This design came with challenges where some blocks were very lossy.

1 Introduction

When an athlete or a soldier gets a concussion, the trainers or staff want to be able to determine quickly whether the person has suffered a traumatic brain injury (TBI) or not. The current methods for detecting a TBI are magnetic resonance imaging (MRI), computed tomography (CAT Scan), or x-ray. The problem with these methods is that the patient is required to be in the hospital. The solution needs to be fast, so that a course of action can be taken quickly, portable, so that trainers can take it onto a sports field or court or so a service member can be evaluated on a battlefield, noninvasive, so that no splitting of the skin or skull is required to determine a result, and it needs to be designed using commercially available components.

The way fNIR spectroscopy works is that it relies on the detection of hemoglobin. Hemoglobin is an iron-rich protein that is responsible for transporting oxygen. Oxygenated hemoglobin travels to the brain and deoxygenated hemoglobin travels out of the brain. In a healthy brain, the ratio between oxygenated and deoxygenated hemoglobin should be about equal. However, when the brain suffers a big impact, small blood vessels can rupture and this allows for hemoglobin to leak. The system uses this fact to detect a difference in the ratio of oxygenated and deoxygenated hemoglobin.

There are three wavelengths that are significant to the design of this project. Around 650-700nm is where the deoxygenated hemoglobin absorbs the near-infrared light most strongly. Around 850nm is where the oxygenated hemoglobin absorbs the near-infrared light most strongly. At about 800nm is called the isosbestic point and this is where oxygenated and deoxygenated hemoglobin absorb the light identically. These three wavelengths can set up a system of three equations using the Modified Beer-Lambert Law and be used for calculations in the project.

The system is not meant to be a replacement of MRI or CAT Scan or any other current technologies used in medical settings for the detection of TBIs. The system is rather supposed to be an option for fast detection for athletes and soldiers.



Figure 1: A conceptual image displaying a system that can wirelessly display tomography of the brain.

2 Background

In functional near-infrared spectroscopy, there are three main approaches for the solution.

2.1 Continuous Wave NIR Spectroscopy

As the name suggests, it uses a continuous or constant intensity. The system measures a change in the light that has passed through the brain tissue. CW systems rely on a few assumptions because the light scattering and absorption coefficients can not be determined absolutely [13].

2.2 Time Domain NIR Spectroscopy

Time Domain utilizes short pulses of light into the brain tissue. The system measures the arrival time distributions of the photons after they have passed through the tissue.

2.3 Frequency Domain NIR Spectroscopy

FD-fNIRS sends intensity modulated near infrared (NIR) light into tissue and gets two quantities from the light that is received by a nearby detector. These quantities are the

amplitude, meaning how much the modulated signal was attenuated, and the phase shift, meaning how much the signal was delayed through the tissue. Because it measures phase as well as amplitude, an FD system can separate the absorption coefficient from the reduced scattering coefficient, which allows the estimation of oxygenated (HbO_2) and deoxygenated (Hb) hemoglobin rather than only relative changes [12].

3 Approach

3.1 General Block Diagram

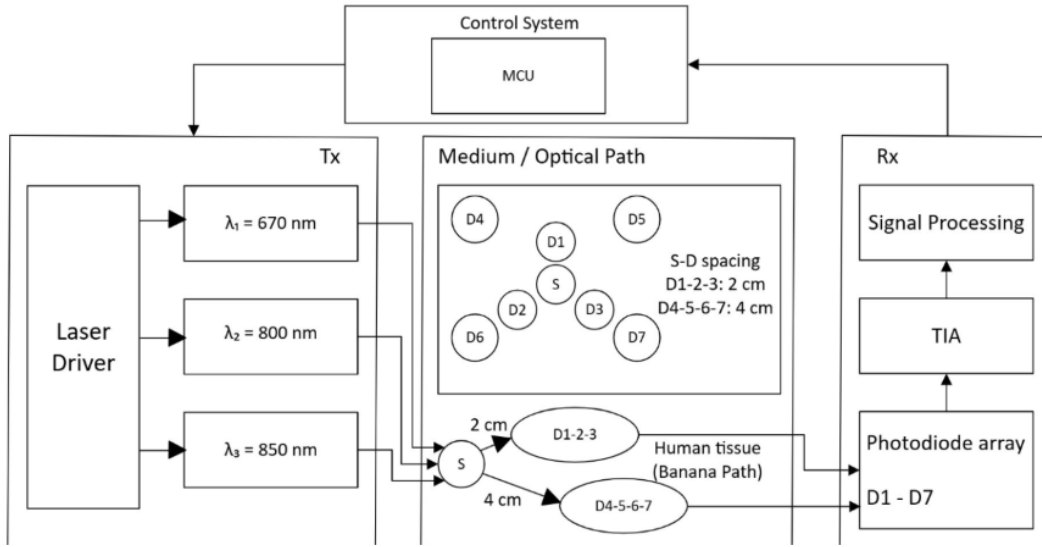


Figure 2: General System Block Diagram.

3.2 Transmitter

Project Overview This section proposed a multi-wavelength optical transmitter system using three VCSEL laser diodes at 680nm, 780nm and 850nm. The purpose for this sub system is to generate a controllable RF-modulated optical output where the system can select one wavelength channel at a time and modulate that laser using a high frequency RF signal. The transmitter system is designed around four main sub systems: a frequency generation block, a DC bias block, a RF modulated VCSEL output stage block and a digital control block.

Frequency generation block: The RF signal is generated by ADF4351 PLL Synthesizer through the digital control of ESP32 (via SPI), amplified by a RF amplifier (SBB5089Z) and eventually routed through an SP3T RF switch.

DC bias/Current stabilizer block: The DC current-controlled path includes three V-I Converters between signals coming out of ESP32 and DAC; within V-I Converters, there exists an Op-Amp, a MOSFET Switch and a sense resistor.

RF modulated VCSEL output stage block: Since there are two distinct RF and DC paths within the system, the signals will be ultimately combined at bias tees before being fed into the laser diodes.

System Objective The goal of this transmitter design is to create a compact optical source that can produce steady frequency modulated light at multiple wavelengths. The system should allow us to satisfy these requirements for the project:

1. Generate a tunable RF signal in the required modulation range.
2. Select which wavelength channel receives RF signal.
3. Independently set the DC bias current to each of the laser diodes.
4. Produce a stable modulated optical output for testing and actual system integration.

This design is useful since each laser diode requires different controlled DC operating current, while the RF signal must be applied separately as a modulation signal. Separating the DC bias path and RF modulation path makes the system modular, easy to tune and develop.

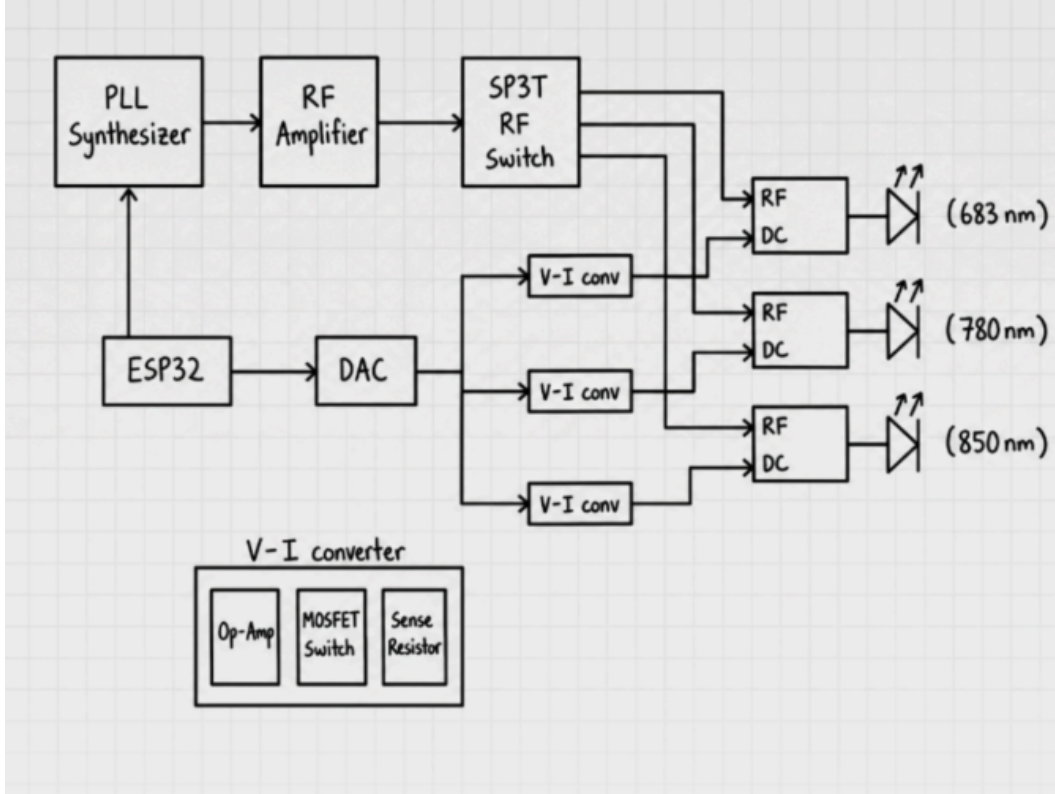


Figure 3: Transmitter System Architecture.

System Architecture and Operation Everything starts from the digital control block where ESP32 acts as the main controller. It communicates with the ADL4351 PLL Synthesizer through SPI (digitally controlled) to set the desired RF frequency, controls the MCP4728 DAC to adjust laser bias currents, and controls the SP3T RF switch to choose which wavelength channel is active.

RF generator block: For the RF generator block, ADF4351 would be a suitable pick since it can generate RF output frequencies from 35MHz to 4.4GHz, which easily covers the desired 300-600MHz modulation range. The RF signal is then passed into the SBB5089Z RF Amplifier, which is a broadband RF gain block operating from 50MHz to 6GHz; these two specs make them both compatible with each other and operate well within our desired frequency range. After amplification, the signal enters the JSW3-272DR+ SP3T RF switch. This switch allows one RF input to be routed to one of three output channels. Since there is only one laser diode that actually operates at a time, this design allows the flexibility where we can control which laser diode receives the RF modulation signal. The JSW3-272DR+ operates from 5MHz to 2.7GHz, so it also covers the 300MHz-600MHz range.

DC current controlled bias block: Similar to the RF modulation path where each laser diode has its own RF branch, it also has its separate DC current-controlled path. The MCP4728 is a quad 12-bit DAC with I2C control which is useful because one chip can provide multiple programmable analog voltages. In this design, three DAC outputs are used

to control three V-I converters, one for each VCSEL. Inside the V-I converters, there exists an op amp, MOSFET and sense resistor to convert voltage into a controlled laser current.

The relationship between controlled voltage, current that laser diodes receive and sense resistor should be:

$$I_{\text{laser}} = \frac{V_{\text{DAC}}}{R_{\text{sense}}} \quad (1)$$

Whereas the sense resistors are chosen based on the safe operating current of each laser diode. For example, if a laser diode requires about 10mA and the control voltage is around 1V, the sense resistor should be 100 Ω in order to produce the desired current value. The LM324 is a quad op amp that is acceptable for the DC bias path since it is only controlling low frequency/DC bias current, not directly handling the high frequency RF modulated signal.

Bias tee: The final stage of each path is the bias tee, shown as the RF/DC block in the diagram. The bias tee combines the DC bias current from the V-I converter as well as the RF modulation signal from the RF switch. In this design, the DC path biases the laser diode above its threshold current, while the RF signal modulates the optical output at a desired frequency. This combination allows the laser diode to emit light whose intensity changes at the selected RF frequency.

Component Compatibility Justification Since the main idea is to separate the DC path and the RF path, each selected component is compatible because each block handles their correct path of the signal chain. As discussed above, ADF4351, SBB5089Z and JSW3-272DR+ all support the targeted frequency range of 300MHz to 600MHz. The RF switch chosen also matches the idea of three-channel structure of the transmitter since it is a SP3T device, meaning one input can be routed to one of three outputs.

The ESP32 is particularly suitable for not only the operation of the transmitter, but also for the receiver as well. It provides digital control interfaces such as SPI/I2C/GPIO. Because of that, the specific choice of ADF4351 is highly compatible for the design since it can directly communicate with ESP32 via SPI. Furthermore, the DAC (MCP4728) also uses I2C. This makes the ESP32 a good central controller for frequency setting and laser bias adjustment.

The MCP4728 DAC fits well because the design needs multiple adjustable analog voltages. Since there are three laser diodes, each diode can have its own DAC-controlled bias setting. This matches the design's philosophy of being modular and having a high flexibility in component choices.

The V-I converter block is essential here because the laser diodes are current-controlled rather than voltage-controlled. In addition, a small fluctuation in voltage could cause a large change in laser current, leading to inaccurate data or potentially affecting the laser diode's operation. Because of that, this specific combination of the op amp, MOSFET and the sense resistor all serve the common purpose of providing the laser diodes stable current at its operating value.

The specific choice for VCSEL laser diodes is appropriate because VCSELs typically have

low operating current, and most importantly, they all have high-speed modulation capability. The 680nm VCSEL datasheet shows a fast 80ps rise/fall time, which is much faster than the required 300MHz-600MHz modulation range. It's also the same for 780nm and 850nm where "High speed 2.5Gbps" is listed as one of the main features of the laser diode.

Proposed Working Mechanism During operation, ESP32 first sets the desired RF frequency to ADF4351. The PLL then outputs a stable signal at an accurate value. Since the signal output from PLL Synthesizer is extremely weak, the signal then has to be amplified by SBB5089Z and ultimately sent to the SP3T RF switch in order to be selected into each laser diode.

At the same time, the ESP32 programs the MCP4728 DAC to set the correct DC bias current for each VCSEL. Each DAC output controls a V-I converter, which supplies a stable controlled current to its corresponding laser diode.

While the SP3T switch selects which channel to feed the RF signal into, the DC bias path also has this capability but with a different mechanism. When one wavelength is selected, the ESP32 sets the selected VCSEL's DAC-controlled bias current to its operating level. The other VCSEL bias currents are set to zero or below threshold so that they remain off.

This means the transmitter does not need three separate RF generators. Instead, one RF source is shared among three optical channels, and the RF switch chooses which laser receives the modulation. By following this approach, it reduces the cost, complexity and also PCB space.

Expected Outcome The expected result would be a functional transmitter prototype that can produce a selectable RF-modulated optical signal at three wavelengths. The system should be able to:

1. Generate RF modulation in the 300MHz-600MHz range, with increments of 100MHz.
2. Select one wavelength channel using the RF switch.
3. Adjust each VCSEL's DC bias current and desired modulation frequency all through ESP32 software.
4. Produce modulated optical output from the selected VCSEL.

The system is designed to be modular, so ideally each block could be tested separately before the full system integration.

Conclusion The proposed transmitter design is practical because each selected component serves a clear purpose and matches the project's requirement. The ADF4351 provides a tunable and digitally controlled RF source, the SBB5089Z amplifies the output RF signal

and JSW3-272DR+ selects one of three wavelength channels. On the other side, MCP4728 and V-I converters provide programmable laser current control. Eventually, two branches meet at the bias tee where it would be fed into the VCSELs.

Overall, the design is compatible at the proposed level since the RF components operate well within the required frequency range. Furthermore, both the RF block and DC block are controlled through ESP32, which makes every operation related to the transmitter programmable.

3.3 Receiver

The receiver (Rx) chain is responsible for four tasks, detecting the weak optical signal and converting it to photocurrent, routing the signals from the seven detectors through a shared measurement path, extracting amplitude and phase in the analog domain, and digitizing only the slow extracted DC values and transmitting them wirelessly.

The central design decision is that performing amplitude and phase extraction in analog hardware means the system only ever has to digitize DC voltages. A receiver that instead digitized the full waveform would have to sample the 500 MHz directly, which requires a very fast analog to digital converter and an FPGA. The analog extraction approach replaces all of that with a 16 bit, 860 SPS I2C ADC and a microcontroller. This is what makes the system portable [1].

Seven S5972 silicon PIN photodiodes, each followed by its own OPA855 transimpedance amplifier (TIA), feed a single PE42512 RF switch. The switch selects one detector at a time and it goes to a TCP-2-10+ two way power splitter. The splitter produces two identical copies of the selected signal. The first copy, the phase path, is boosted by the LTC6431-15 RF gain block and then compared against the modulation reference (REF Frequency) in the AD8302 phase detector. The second copy, the amplitude path, goes directly to the AD8307 amplitude detector. The DC outputs of both detectors are digitized by the ADS1115 16 bit ADC and read by the ESP32 microcontroller.

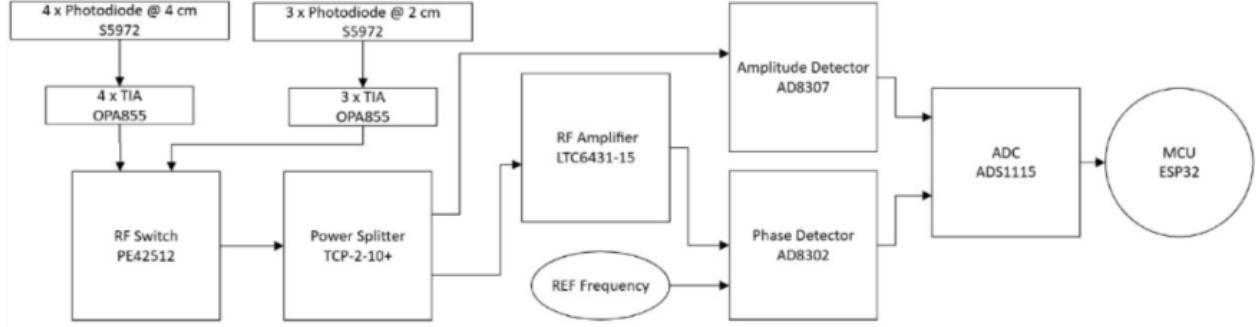


Figure 4: Receiver (Rx) Block Diagram. Seven S5972 photodiodes (three at a 2 cm source detector separation, four at 4 cm) are amplified by an OPA855 TIA, multiplexed by the PE42512 RF switch, split by the TCP-2-10+, and processed by two parallel detectors which are the AD8302 (phase, fed by the LTC6431-15 gain block and the modulation reference) and the AD8307 (amplitude). The ADS1115 digitizes both DC outputs for the ESP32.

The photodiodes are placed at two source detector separations. There are three at 2 cm and four at 4 cm. The depth of tissue penetrated by the diffused NIR light scales with the separation. The photon path forms a banana shaped region that reaches a depth of about half the source detector distance and sensitivity to deeper brain tissue increases as the separation grows from about 2 cm and up [11, 12]. The shorter 2 cm channels are weighted toward scalp and skull tissue, while the 4 cm channels reach further where a vascular leak from a concussion or TBI would appear. The present system uses seven detectors at 2 cm and 4 cm to go deeper into the tissue and to average over several positions.

A phase measurement is only meaningful relative to a reference. The REF Frequency input to the AD8302 is a copy of the modulation frequency of 500 MHz, that intensity modulates the laser diodes on the transmit side. The AD8302 gives the phase difference between the tissue delayed signal and the reference. That difference is the FD-fNIRS phase observable, which is proportional to the time of flight through the tissue [7].

Each S5972 converts the NIR optical power that goes through the tissue path into a photocurrent. The S5972 is a high speed Si PIN photodiode with a cutoff frequency of 500 MHz at a reverse bias of 10 V. It has a spectral response spanning about 320 to 1000 nm with a peak near 800 nm [2]. These properties of bandwidth and spectral coverage aid the design. A 500 MHz cutoff matches the 500 MHz modulation frequency, so the diode can reproduce the modulation envelope that carries the phase information. All three operating wavelengths, 670, 800, and 850 nm, fall in the high responsivity region of the diode. The responsivity values used are about 0.46, 0.54, and 0.54 A/W, which peaks near 800 nm. Since it is a PIN rather than avalanche diode it has no internal gain.

The OPA855 is a wideband, low noise, operational amplifier that is specifically for wideband transimpedance applications and for weak signal inputs [3]. The OPA855 converts each photodiode's nanoampere level current into a voltage across its feedback resistor. With a 1 k Ω feedback resistor the resulting TIA outputs are a few hundred microvolts. The gain

bandwidth and low noise support the design. An 8 GHz gain bandwidth product supports a TIA bandwidth above the 500 MHz carrier. An input voltage noise of $0.98 \text{ nV}/\sqrt{\text{Hz}}$ keeps the amplifier from setting the receiver noise floor.

The OPA855 is stable at a gain of seven. This is an advantage because it gives more bandwidth for the same power [3]. The block diagram gives each photodiode its own OPA855 rather than switching at the photocurrent level so that each weak photocurrent is amplified at the diode before any lossy RF routing which is what preserves the signal to noise ratio.

The PE42512 takes one of the seven amplified detector signals at a time and sends it to the detector chain. One amplitude and phase detection chain is needed for all seven detectors instead of replicating the splitter, gain block, and two detectors seven times. The seven inputs use seven of the twelve throws and leave five spares for future channels. 9 kHz to 8 GHz covers the 500 MHz frequency. A 232 ns switching time allows the ESP32 to cycle through all detectors many times within the 10 Hz refresh budget. The datasheet gives an insertion loss of 1.3 dB for the switch at 6 GHz but the real loss at 500 MHz would be lower.

The TCP-2-10+ splits the selected detector signal into two identical copies. This allows amplitude and phase to be measured at the same time and from the same signal. One copy goes to the phase path, the LTC6431-15 and then the AD8302, and the other to the amplitude path, the AD8307. It covers 5 to 1000 MHz, with a total loss of about 3.1 to 3.3 dB, the inherent 3 dB split plus a small insertion loss, and very low amplitude unbalance, about 0.14 dB, and phase unbalance, under 0.15° [5].

The LTC6431-15 adds 15.5 dB of gain in the phase path only, raising the signal into the usable input window of the AD8302. The specified input range of the AD8302 is -60 dBm to 0 dBm [7]. The raw signal sits about 10 dB below the AD8302 floor and would not be measured reliably. Adding the 15.5 dB of gain from the LTC6431-15 raises the phase path to about -56.0, -53.9, and -57.3 dBm, all inside the -60 to 0 dBm window. The LTC6431-15 delivers 15.5 dB of power gain over 20 to 1700 MHz and has a low noise figure of 3.33 dB which means it doesn't significantly impact the signal to noise ratio [6].

The AD8302 measures the phase difference between the returning signal and the modulation reference and it can also measure the gain ratio between them, presenting each as a DC voltage, phase at 10 mV/degree and gain at 30 mV/dB. Phase is the quantity that gives FD-fNIRS its advantage because it encodes the time of flight of the modulation envelope and therefore the effective optical pathlength and that allows for absorption and scattering to be separated [12]. The extracted phase is effectively a clean DC value.

The AD8307 is a logarithmic amplifier covering DC to 500 MHz with a 92 dB dynamic range, a slope of 25 mV/dB, and a log intercept near -84 dBm [8]. It measures the amplitude, the RF power, of the returning signal and outputs a DC voltage proportional to power in dB. The basic range of the AD8307 reaches down to about -75 dBm and to about -88 dBm with an input matching network. The amplitude path levels of -71.5, -69.4, and -72.8 dBm are above this floor, which is why the amplitude path needs no preamplifier and why the LTC6431-15 was added to the phase path only. It has a 25 mV/dB slope and a -84 dBm intercept so the outputs are about 0.313, 0.365, and 0.280 V, which is in the ADS1115 input

range.

The ADS1115 is a precision, low power, 16 bit delta sigma ADC with four single ended, or two differential, input channels, a data rate up to 860 samples per second, a programmable gain amplifier with full scale ranges from ± 256 mV to ± 6.144 V, an internal reference and oscillator, and an I2C interface [9]. The ADS1115 digitizes the DC outputs of the AD8302, phase and optionally gain, and the AD8307, amplitude, and delivers them to the ESP32 over I2C. Because amplitude and phase have already been reduced to DC, the system never samples the 500 MHz carrier. An 860 samples per second ADC is more than the 10 Hz refresh rate needed, replacing the hundreds of MSPS converter and the about 2.4 Gb/s data link that a digitize everything front end would require [1]. Four channels accommodate the phase, and gain, outputs of the AD8302 plus the amplitude output of the AD8307.

The ESP32, specifically the ESP32-WROOM-32 runs the receiver. It sets the PE42512 to select a detector, reads the ADS1115 over I2C, executes the data processing, the optical density change and the MBLL solve for the changes in HbO_2 and Hb and sends the results over Wi-Fi at the 10 Hz refresh rate.

The received power is equal to the transmitter output power plus the gain of the transmitter antenna minus transmitter losses minus the path loss minus any other losses plus the receiver antenna gain minus receiver losses. Since the antennas are built into the ESP32 and the laptop, the transmitter and receiver losses are dropped. In the case of an athlete getting hurt on the field, the laptop and device can stay on the sidelines if the athlete is capable of making it there or of being assisted there or the laptop and device could be brought out onto the field in a more serious situation with stoppage of play. With this in mind, the path length is going to be assumed to be 1-4 meters. The transmitting power of the ESP32 is 19.5 dBm [14]. The antennas are assumed to have a +2dBi gain because they are going to be low gain. For the loss of the free space and the body shadowing loss, the following figure from an analysis of human body shadowing effect is used and shows a channel loss of about 60 dBm [15].

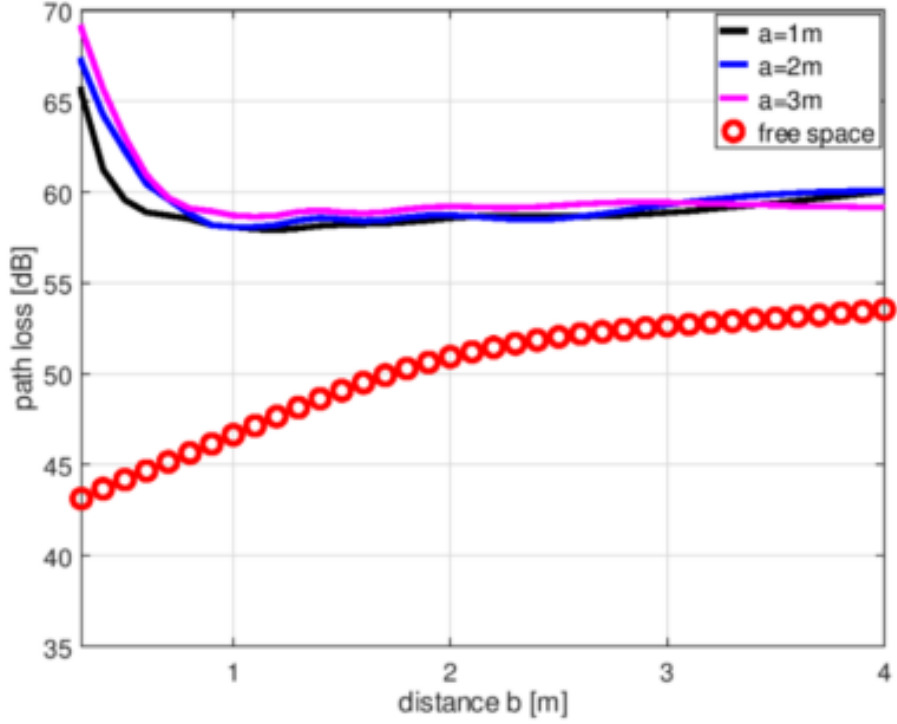


Figure 5: Path loss between transmitting and receiving antenna accounting for body shadowing for 2.4 GHz Wi-Fi.

The received power link budget is calculated as:

$$P_{rx} = P_{tx} + G_{tx} - L_{channel} + G_{rx}$$

$$P_{rx} = 19.5 \text{ dBm} + 2 \text{ dBi} - 60 \text{ dB} + 2 \text{ dBi} = -36.5 \text{ dBm}$$

This value is a strong Wi-Fi signal as the sensitivity floor can typically go down to -90 dBm [16]. This ultimately achieves the goal of the receiver with having the data wirelessly transmitted to the laptop. Ultimately, the near infrared light has been received by the computer as bits.

4 Conclusion

The system is sufficient for the detection of a TBI and can provide helpful information while being faster than other methods. However, this does not intend to replace any of the existing methods. The system has a specific focus on injuries involving athletes and those who serve in the military. In those regards, the system meets the expectation although the system has yet to have been realized as it is still just a design.

The system could be expanded on in multiple ways. With 5G Wi-Fi becoming more available and accessible in the current world, a system that utilizes a microcontroller capable of sending

signals at that frequency could make this a much stronger design. This could evolve into a system that is capable at much larger distances that can account for much larger loss within the channel. Additionally, the microcontroller, the ESP-WROOM-32 advises to use newer designs on the datasheet. The RF switch, the PE42512, has additional throws on it. This means that more photodiodes and operational amplifiers could be added to a future design.

The impact of this system has the potential to save lives. For instance, when someone gets a concussion and would not have otherwise taken the injury as serious enough, the system could notify them of the TBI which would prompt them to seek further medical advice and potentially also treatment. The treatment received could drastically increase the quality of life rather than having someone live with a TBI and not know about it. Alternatively, in cases where the injury may have appeared to be a lot more substantial, the system could notify the staff or service members that a TBI was not detected. While this would not ultimately be the final say in whether or not the person should not need treatment, it could reduce the stress and panic of members of the team and those surrounding the incident. This could lead to a safer environment for staff to work and for crowds at games to be more informed due to faster reporting because of the system. Overall, the FD-fNIRS system that was designed achieved the goals and has lots of room for improvement and refinement.

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